



ASSIGNMENT - 01

Python Programming

DATA SCIENCE

Assignment - 01

Python Programming

PYTHON QUESTIONS

1. What is Python and why is it popular in data science?
2. What are lists and tuples in Python? How do they differ?
3. How are dictionaries different from lists in Python? Give an example use case for a dictionary.
4. Explain what a set is in Python and one advantage of using a set.
5. What is a list comprehension in Python? Write a simple example.
6. What is a lambda function in Python? How is it typically used?
7. How do you handle exceptions in Python? Give a brief example using try/except.
8. What is the purpose of the self keyword in Python class methods?
9. What is the __init__ method in a Python class?
10. What is the difference between == and is in Python?
11. How can you copy a list in Python? What is the difference between a shallow copy and a deep copy?
12. What are *args and **kwargs in Python functions?
13. What is a generator in Python and how do you create one?
14. What is a decorator in Python and what is it used for?
15. How do you read a file in Python and ensure it gets closed properly?
16. What is the Global Interpreter Lock (GIL) in Python?
17. Explain one key difference between Python 2 and Python 3.
18. What are modules and packages in Python? How do you import a module?
19. What is pip and how do you use it?
20. What are NumPy and Pandas, and why are they important for data science?
21. How do you convert a list of strings to a list of integers in Python? (Provide a brief example code.)
22. What does the with statement do in Python? Give an example of using it.
23. How would you merge two dictionaries in Python (in recent versions of Python)?
24. How do you iterate over the key-value pairs of a dictionary?
25. What is list slicing? Give an example of slicing a list.
26. How can you reverse a list in Python?
27. How do you check if a substring exists in a given string in Python?
28. What is a Python virtual environment and why would you use one?
29. How do you write a function in Python that takes an arbitrary number of arguments?
30. What is object-oriented programming (OOP) in Python? Name the four main principles of OOP.